Statistical Investigation & Bias Report

1. Selection Bias (Contactability Check)

- **Finding:** The average recovery rate between fully contactable borrowers and uncontactable borrowers is practically indistinguishable (~23.5% vs ~24.0%).
- **Conclusion:** The business did not artificially inflate its conversion metrics by quietly dropping hard-to-reach borrowers from the active queue. Selection bias is ruled out as the driver of the reported 11% improvement.

2. Mix Effects & Simpson's Paradox (Risk Segment Check)

- **Finding:** The month-over-month distribution of incoming accounts across `HIGH`, `LOW`, `MEDIUM`, and `UNKNOWN` risk segments is locked at a mathematically perfect 25% split for every single month.
- **Conclusion:** In reality, portfolio risk fluctuates based on macroeconomic factors and underwriting changes. This perfectly uniform distribution proves the `risk_segment` data is mechanically generated. The March anomaly was definitively not caused by the business suddenly acquiring "safer, easier-to-collect" debt, as the underlying population demographics never actually changed.

3. Survivorship Bias (Initial Characteristics Check)

- **Finding:** When comparing accounts that successfully recovered against those that failed, the starting parameters are identical. Both cohorts share an exact average principal amount of ~₹402,600 and an average Days Past Due (DPD) of ~56 at the time of account opening.
- **Conclusion:** Real collections portfolios exhibit variance (e.g., smaller balances often cure faster). The identical starting characteristics across both success and failure groups prove the system is not dropping difficult accounts from the denominator to make performance look better. Instead, it exposes yet another layer of synthetic, perfectly balanced dummy data in the core systems.

4. Time-Series & Cohort Effects (Longitudinal Trends)

- **Finding:** The longitudinal tracking of both vendor call dispositions and internal account statuses reveals flat, overlapping, synthetic trends from January through July, followed by a massive 80% volume cliff in August.
- **Conclusion:** The August collapse is not an operational failure; it is a textbook "late-arriving data bias" (cohort immaturity). Recent accounts simply haven't existed long enough to generate interactions or payments. Meanwhile, the flat time-series preceding August confirms the 11% month-on-month improvement in March was a random seasonal fluctuation within synthetic noise, not a sustained operational trend.